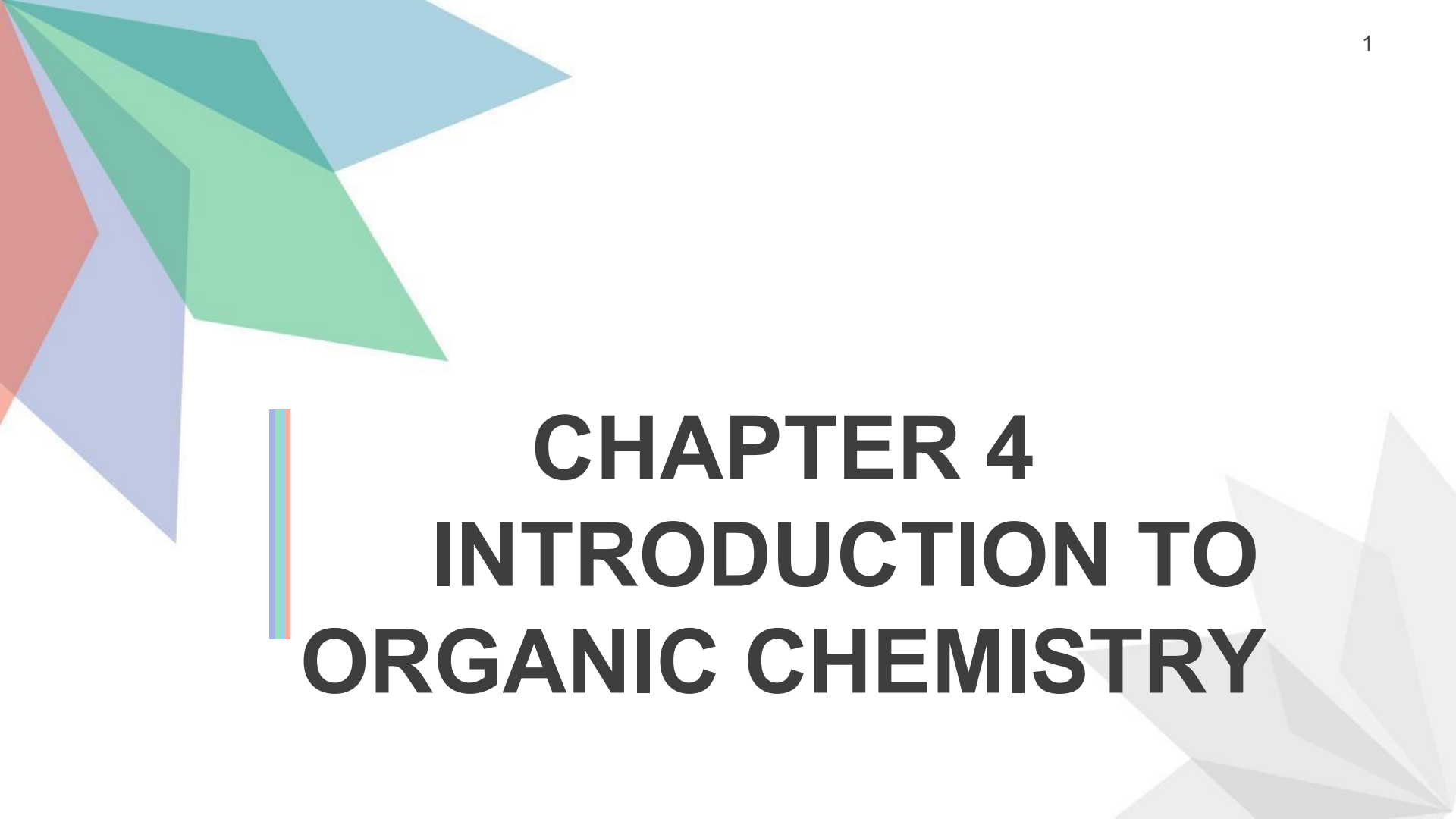


The slide features abstract geometric shapes in the corners. The top-left corner is filled with overlapping triangles in shades of blue, green, and red. The bottom-right corner contains a cluster of overlapping triangles in various shades of gray.

CHAPTER 4

INTRODUCTION TO

ORGANIC CHEMISTRY

4.1 : MOLECULAR AND STRUCTURAL FORMULAE

2

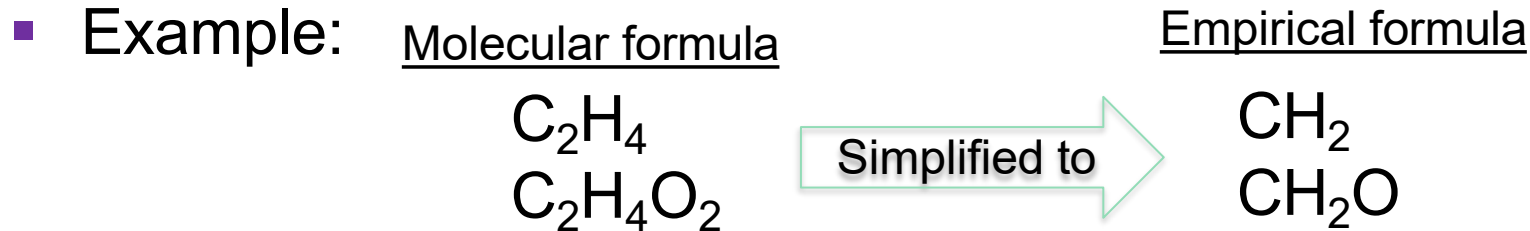
MOLECULAR FORMULA

- Shows the actual number of atoms for all elements present in the molecule.
- Example:



EMPIRICAL FORMULA

- Shows the simplest ratio of atoms for all elements present in the molecule.



STRUCTURAL FORMULA

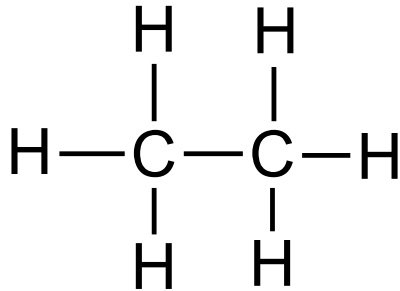
A chemical formula which shows how the atoms in a molecule are bonded to each other.

Has 3 forms :

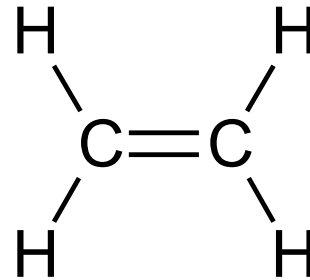
- expanded structure
- condensed structure
- skeletal structure

EXPANDED STRUCTURE

- Shows all the atoms and covalent bonds between atoms.
- Does not represent the actual shapes of the molecules.



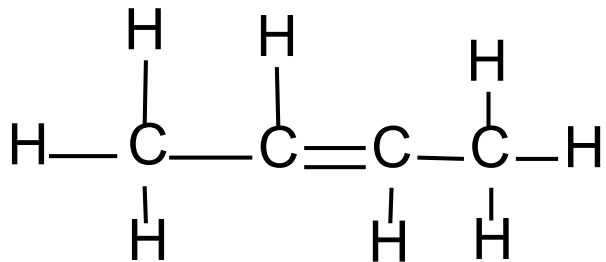
Ethane, CH_3CH_3



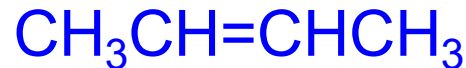
Ethene, CH_2CH_2

CONDENSED STRUCTURE

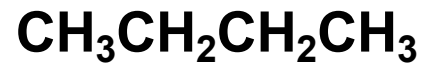
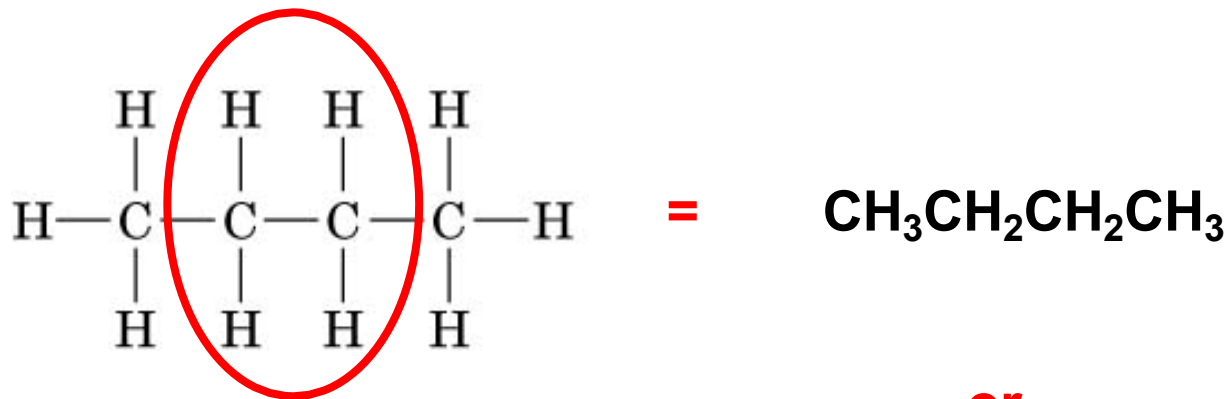
- Does not show C-H and C-C single bonds BUT double and triple bonds are shown.
- All atoms which attached to a particular C are written immediately after the C.



condensed to



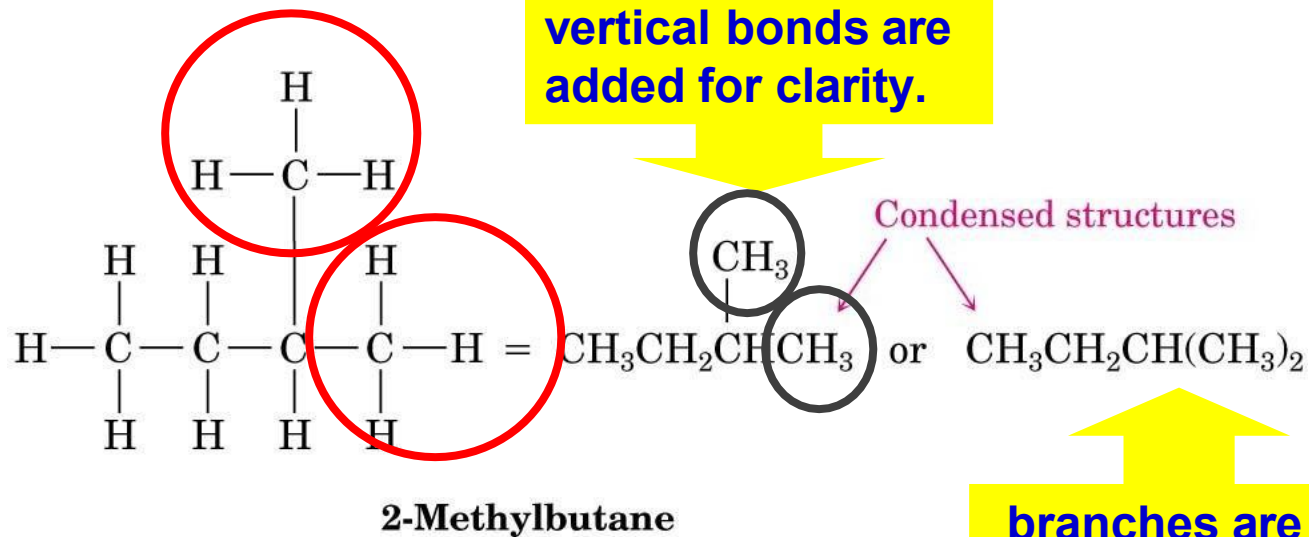
Example



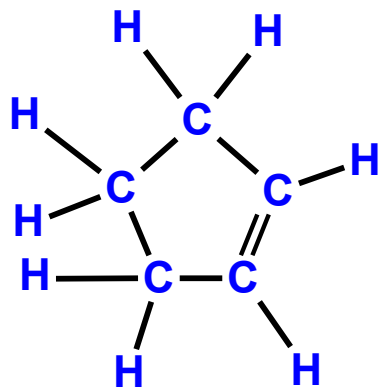
or



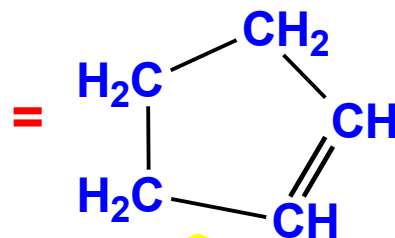
Example



Expanded structure

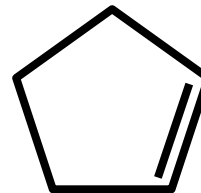


Condensed Structure



**C–C bonds in
a ring must be
shown.**

Skeletal Structure

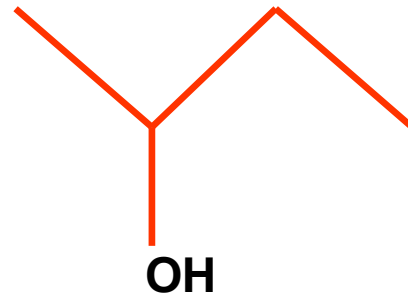


SKELETAL STRUCTURE

- Shows only the carbon skeleton.
- Bonds are represented as lines.
- C atoms are assumed to be at the beginning and end of a line.
- Does not show C and H atoms **but** functional groups and atoms other than C and H **must** be shown.

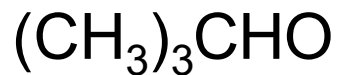
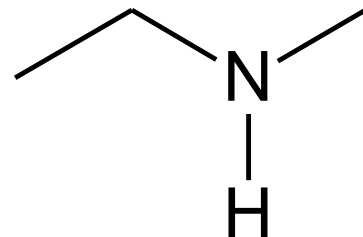


Transformed into
skeletal

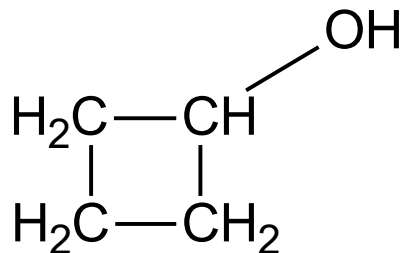
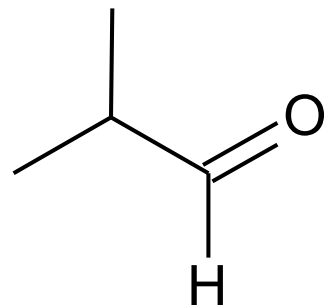




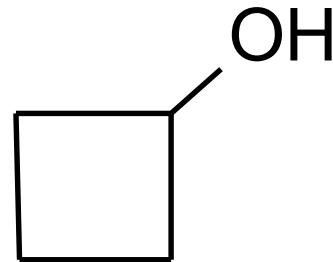
Transformed into
skeletal



Transformed into
skeletal

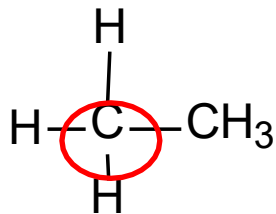


Transformed into
skeletal

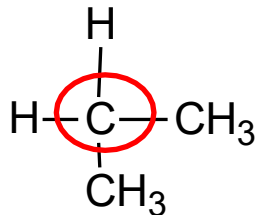


CLASS OF CARBON

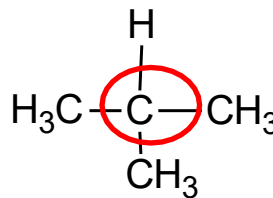
- A primary carbon is bonded to 1 other C atom
- A secondary carbon is bonded to 2 other C atoms
- A tertiary carbon is bonded to 3 other C atoms
- A quarternary carbon is bonded to 4 other C atoms



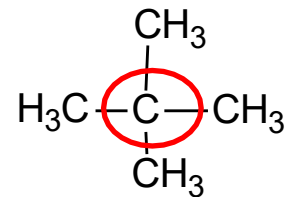
Primary C



Secondary C

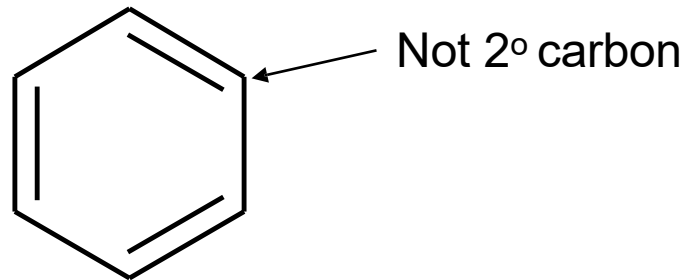
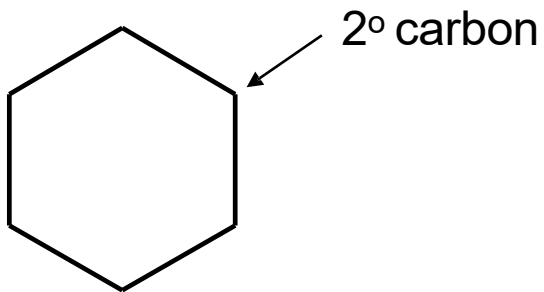


Tertiary C



Quarternary C

Note: Classification of carbon atoms only limited to sp^3 hybridised carbon atoms



Exercise:

Label and state the number of primary, secondary and tertiary carbon.

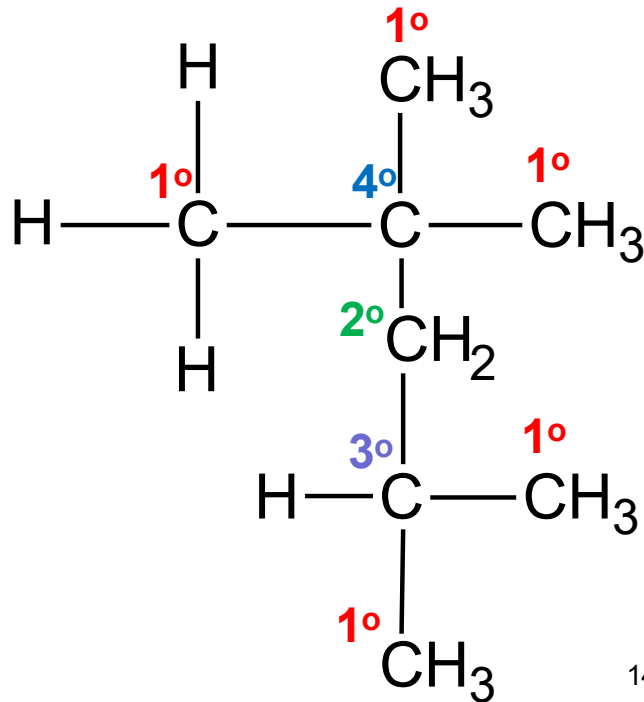
Answer:

1° carbon - 5

2° carbon - 1

3° carbon - 1

4° carbon - 1



CLASS OF HYDROGEN

- A hydrogen atom can be classified according to the class of C attached to it.
- Primary H is bonded to primary C.
- Secondary H is bonded to secondary C.
- Tertiary H is bonded to tertiary C.

4.2 FUNCTIONAL GROUPS & HOMOLOGOUS SERIES

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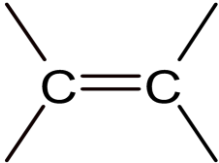
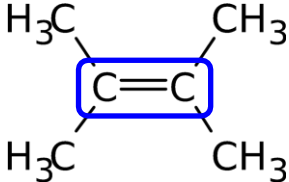
FUNCTIONAL GROUPS

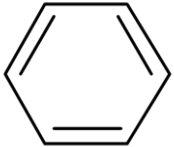
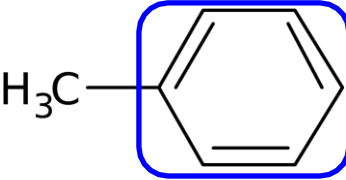
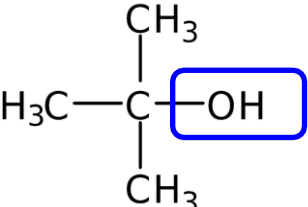
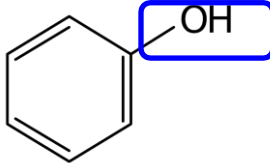
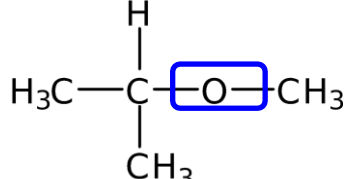
Definition :

- Atom or group of atoms in an organic molecule which characterise the molecule and determine the chemical properties of organic compound.

The importance of functional groups :

- i. The sites where chemical reaction takes place.
- ii. The units by which organic compounds are divided into classes.
- iii. Serve as a basic for naming the organic compounds.

Class	Name of Functional Group	Structure of Functional Group	Example
Alkane	-	-	$\begin{array}{c} \text{CH}_3 \\ \\ \text{H}_3\text{C}-\text{C}-\text{CH}_3 \\ \\ \text{CH}_3 \end{array}$
Alkene	Carbon-carbon double bond		
Alkyne	Carbon-carbon triple bond	$\text{—C}\equiv\text{C—}$	$\text{H}_3\text{C}-\boxed{\text{C}\equiv\text{C}}-\text{CH}_3$

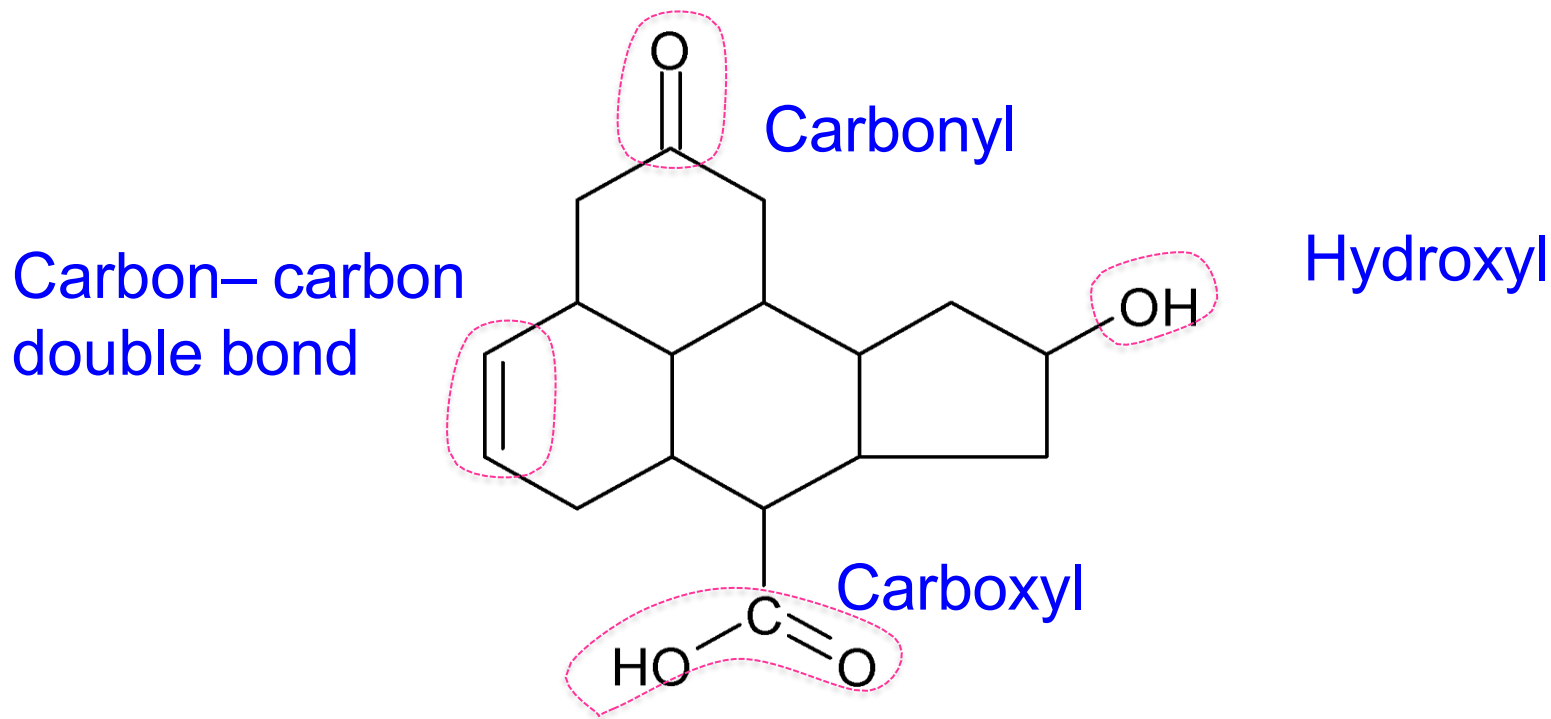
Class	Name of Functional Group	Structure of Functional Group	Example
Arene	Benzene ring		
Alcohol	Hydroxyl	-OH	
Alcohol (Phenol)	Hydroxyl	-OH	
Ether	Alkoxy	-O-	

Class	Name of Functional Group	Structure of Functional Group	Example
Aldehyde	Carbonyl	$\begin{array}{c} \text{O} \\ \parallel \\ \text{---C---H} \end{array}$	$\text{H}_3\text{C---}\boxed{\begin{array}{c} \text{O} \\ \parallel \\ \text{C} \end{array}}\text{---H}$
Ketone	Carbonyl	$\begin{array}{c} \text{O} \\ \parallel \\ \text{---C---R} \end{array}$	$\text{H}_3\text{C---}\boxed{\begin{array}{c} \text{O} \\ \parallel \\ \text{C} \end{array}}\text{---CH}_3$
Carboxylic Acid	Carboxyl	$\begin{array}{c} \text{O} \\ \parallel \\ \text{---C---OH} \end{array}$	$\text{H}_3\text{C---}\boxed{\begin{array}{c} \text{O} \\ \parallel \\ \text{C} \end{array}}\text{---OH}$
Ester	Carboalkoxy	$\begin{array}{c} \text{O} \\ \parallel \\ \text{---C---OR} \end{array}$	$\text{H}_3\text{C---}\boxed{\begin{array}{c} \text{O} \\ \parallel \\ \text{C} \end{array}}\text{---OCH}_3$

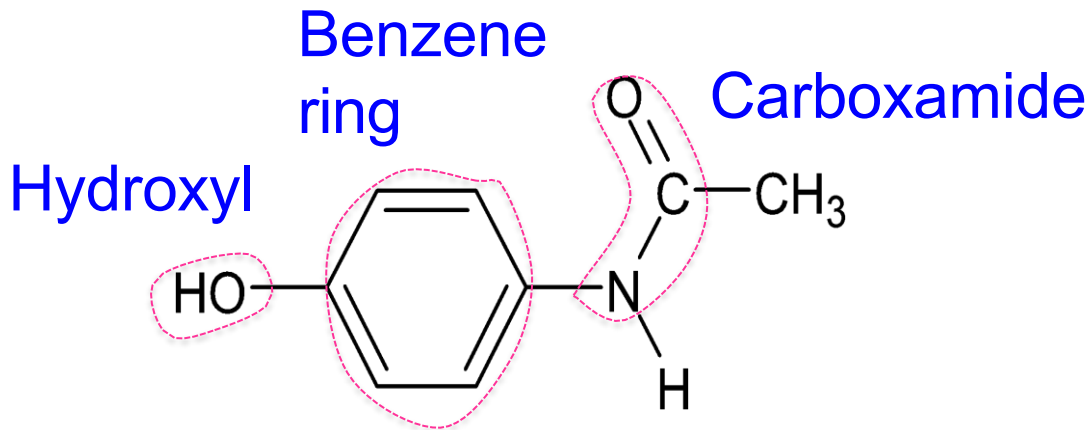
Class	Name of Functional Group	Structure of Functional Group	Example
Amide	Carboxamide	$\text{—}\overset{\text{O}}{\parallel}\text{C—NH}_2$	$\text{H}_3\text{C—}\overset{\text{O}}{\parallel}\text{C—NH}_2$
Amine	Amino	—NH_2	$\text{H}_3\text{C—CH}_2\text{NH}_2$
Nitrile	Cyano	$\text{—C}\equiv\text{N}$	$\text{H}_3\text{C—C}\equiv\text{N}$
Haloalkane	Halogen	F @ Cl @ Br @ I	$\text{H}_3\text{C—CH}_2\text{Cl}$

Exercise:

1. Redraw the following structure, then circle and name all the functional groups. [4 marks]



2. The structure of paracetamol is shown below. Redraw the²² structure, then circle and name all the functional groups that exist. [3 marks]



HOMOLOGOUS SERIES

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Definition : a series of compounds where each member differs from the next member by a $\text{--CH}_2\text{--}$ group.

Example:	CH_3OH	$\text{CH}_3\text{CH}_2\text{OH}$	$\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
	methanol	ethanol	propanol

CHARACTERISTICS of HOMOLOGOUS SERIES

1. All the members of a particular homologous series share the same general formula.
2. All the members of a particular homologous series have the same functional group and same chemical properties.
3. The successive members in the homologous series differs by a -CH_2 group

4. Members of the homologous series show gradual change in²⁵
physical properties as molar mass increases.

Example of homologous series:

General formula: $C_nH_{2n+2}O$

Name	Structural formula	Molecular formula	Molar mass (g/mol)	Boiling point (°C)
Methanol	CH ₃ OH	CH ₄ O	32	64.7
Ethanol	CH ₃ CH ₂ OH	C ₂ H ₆ O	46	78.3
1-propanol	CH ₃ CH ₂ CH ₂ OH	C ₃ H ₈ O	60	97.1

4.3 ISOMERISM

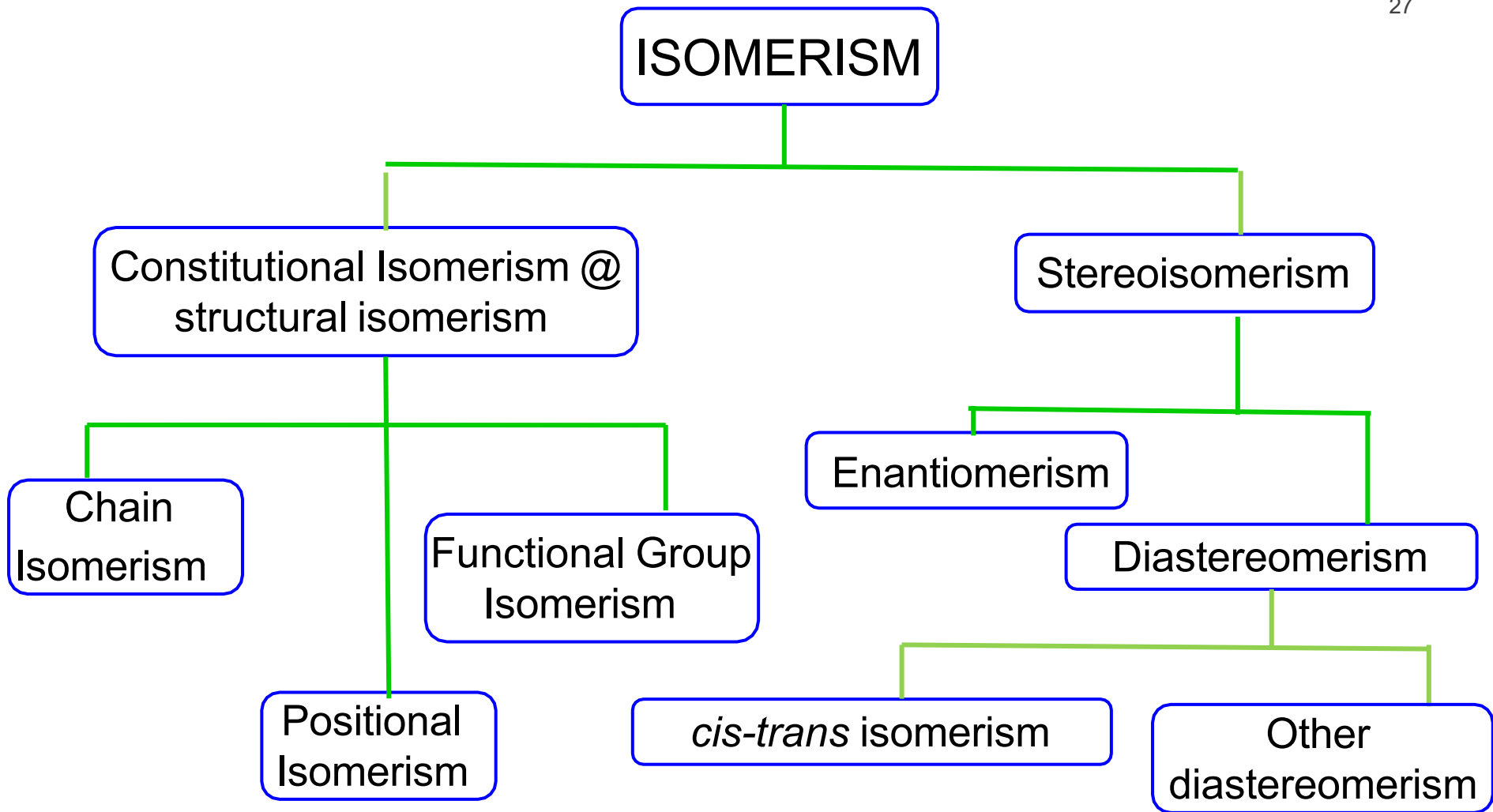
26

ISOMERISM

Definition : is the existence of different compounds with the same molecular formula.

ISOMERS

Definition : different compounds with the same molecular formula.



Constitutional Isomerism

- Existence of different compounds having the same molecular formula but differ in the way the atoms are connected to each other.
- There are 3 type of constitutional isomers:
 - a) Chain isomer
 - b) Positional isomer
 - c) Functional group isomer

CHAIN ISOMERS

- Chain isomers differ in the carbon skeleton.
- can be either straight chain or branch chain.

CHARACTERISTICS OF CHAIN ISOMERS

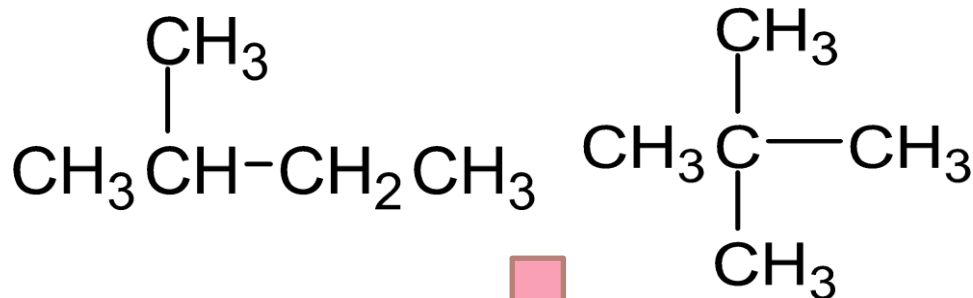
30

- possess the same functional group.
- belongs to the same homologous series.
- similar chemical properties but different physical properties.

Example:



Straight chain



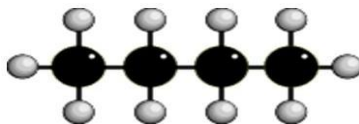
Chain with branching

PHYSICAL PROPERTIES OF CHAIN ISOMERS ³¹

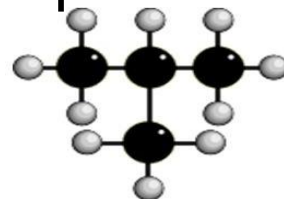
Properties such as boiling point show trends according to the degree of branching.

- Straight chain/linear form isomers have higher boiling points than branched chain isomers.
- The greater the degree of branching, the lower the boiling point.
- Branching decreases the effectiveness of intermolecular attractive forces, so less energy is required to separate the molecules.

Example:



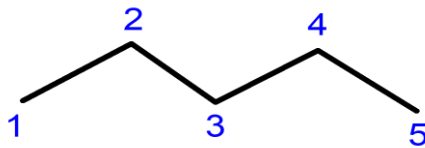
Boiling point = - 0.5°C



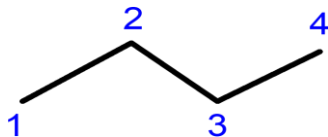
Boiling point = - 11.7°C

THE FOUR STEPS FOR DRAWING CHAIN ISOMERS:

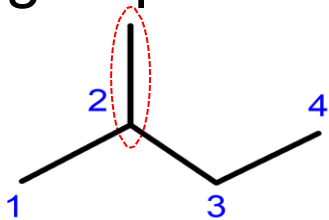
1. Draw the main chain (the straight chain containing all the C)



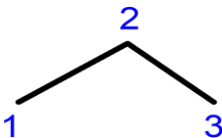
2. Draw the main chain minus 1 carbon



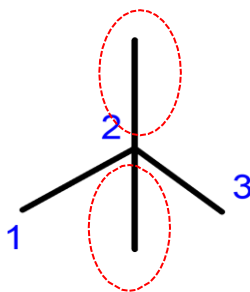
3. Add a methyl group to as many positions as possible. Never add the methyl groups to the end of the chain.



4. Draw the main chain minus 2 carbons.



And add 2 methyl groups to as many positions as possible.



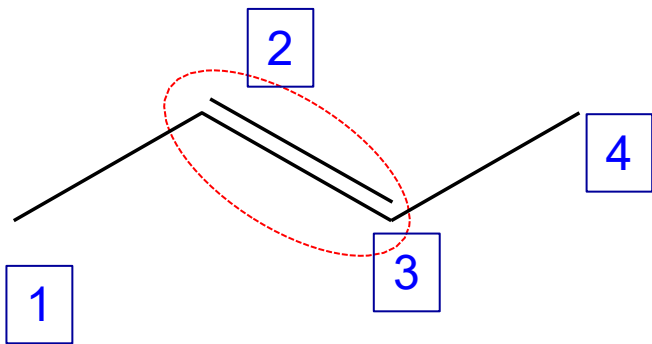
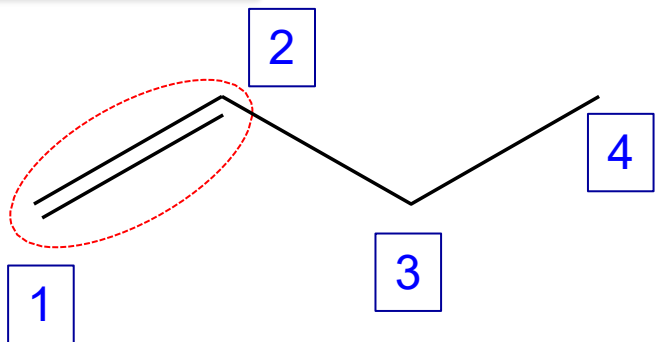
Continue subtracting and adding groups in this fashion until you run out of carbons

POSITIONAL ISOMERS

- Positional isomerism occur in molecules which contain same functional group.
- Positional isomers are different molecules having the same carbon skeleton but differ in the position of the functional group.

Example:

For the following molecules:

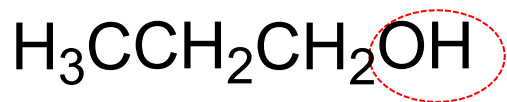


- The molecular formula remains the same, C_4H_8 .
- The types of functional group also remains the same.
- Position of functional group in the molecule changes.

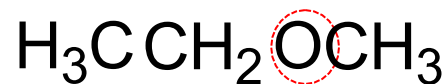
FUNCTIONAL GROUP ISOMERS

- Functional group isomers are molecules with same molecular formula but different functional group.
- Isomers belongs to different homologous series.
- Isomers have different chemical properties and physical properties.

Example:



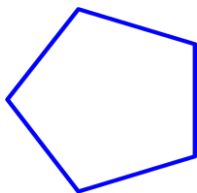
Alcohol has the **hydroxyl group, -OH**



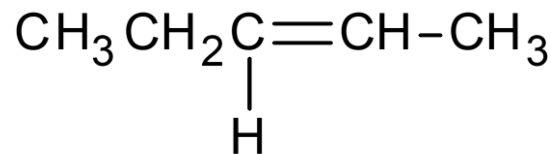
Ether has the **alkoxy functional group**

Pairs of compounds which exhibit functional group isomerism.

1. Cycloalkane and alkene (molecular formula : C_nH_{2n})

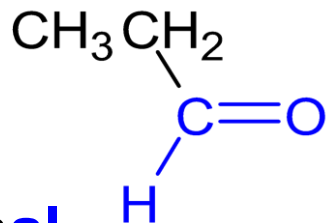


Cyclopentane

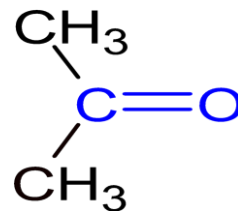


2-pentene

2. Aldehyde and ketone (molecular formula : $C_nH_{2n}O$)



Propanal

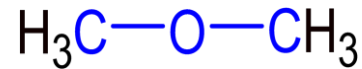


Propanone

3. Alcohol and ether (molecular formula : $C_nH_{2n+2}O$)

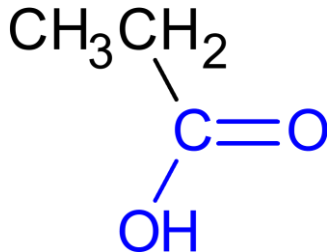


ethanol

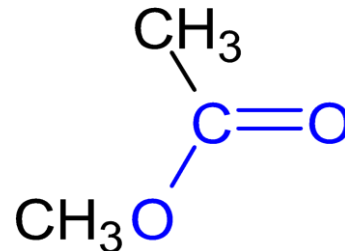


methoxy methane

4. Carboxylic acid and ester (molecular formula : $C_nH_{2n}O_2$)



propanoic acid

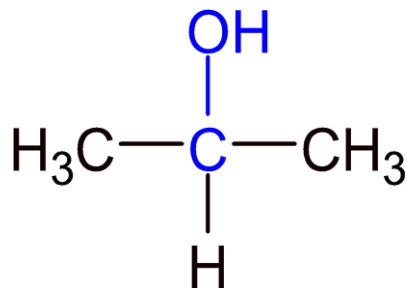


methyl ethanoate

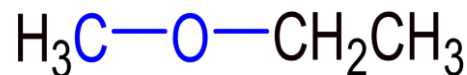
Example:

A compound X has a molecular formula of $\text{C}_3\text{H}_8\text{O}$. Draw and name two possible structural formulae of X with different functional groups.

Answer:



2-propanol



methoxy ethane

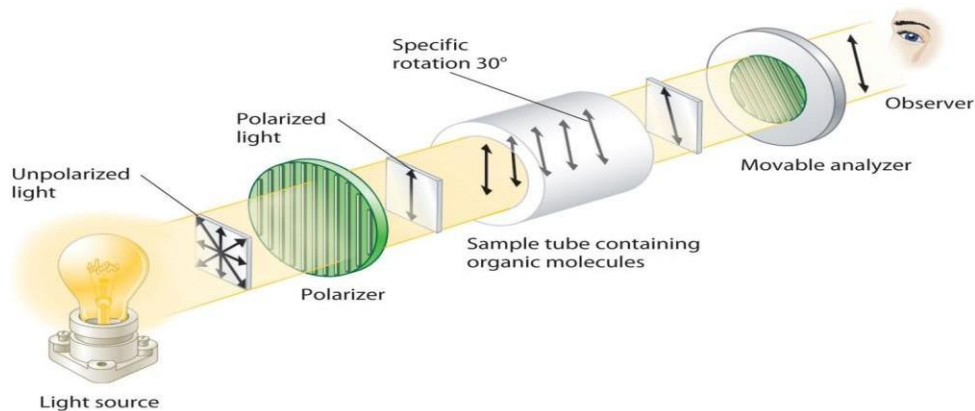
STEREISOMERISM

- The existence of different compounds having the same molecular formula and structural formula but differ in the arrangement of atoms or group of atoms in space.
- 2 types of stereoisomers:
 - Enantiomers
 - Diastereomers

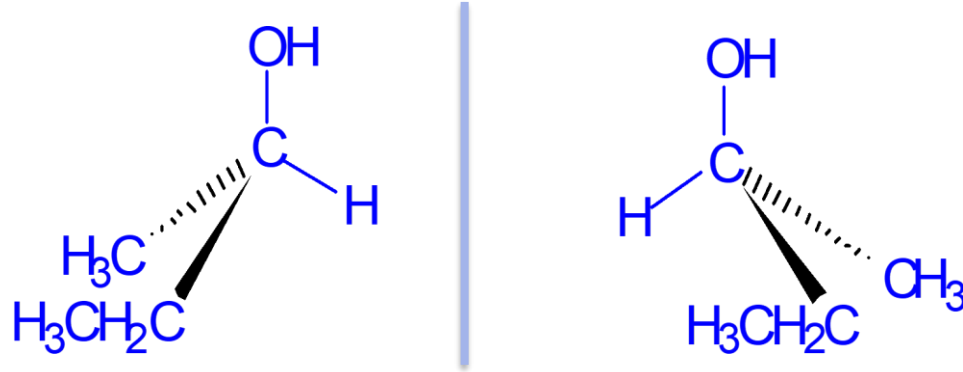
ENANTIOMERS

- Definition: Enantiomers are a pair of stereoisomers that are mirror images but non-superimposable on each other.
- Requirement to be an enantiomer:
 1. There is at least **one chiral carbon** in the molecule.
 2. The molecule and its mirror image are **not superimposable**.
- **Chiral carbon** is a sp^3 hybridised carbon atom with 4 different atoms or groups attached to it.

- Enantiomers are optically active.
- Optically active means able to rotate the plane polarised light.
- The optical activity of a molecule is measured using a polarimeter.



- A pair of enantiomers have same chemical and physical properties but differ in the direction of rotation of the plane polarized light.

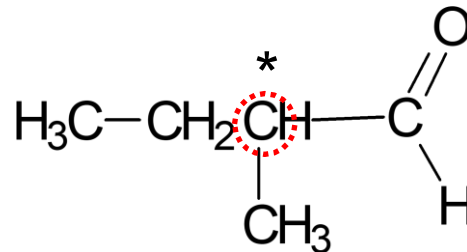


enantiomers

Steps To Draw Enantiomers

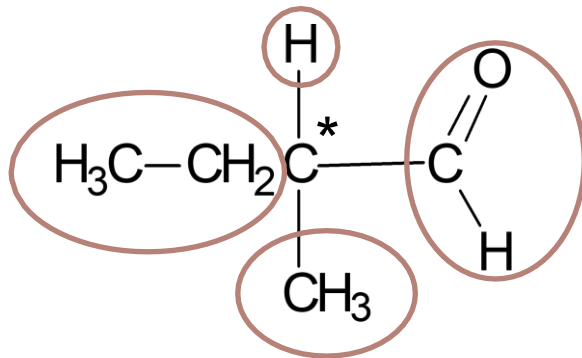
1

Identify chiral carbon and label the chiral carbon with asterisk (*)



2

Identify 4 different groups bonded to chiral C.



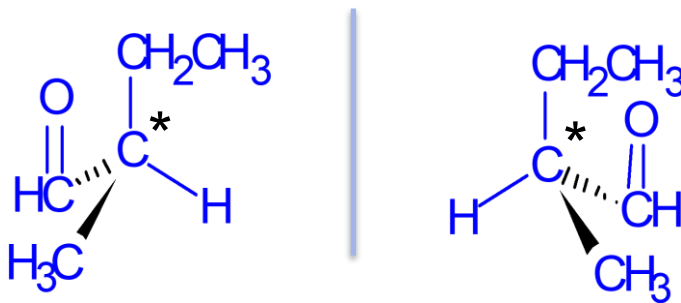
3

Transform into 3D as follows:



4

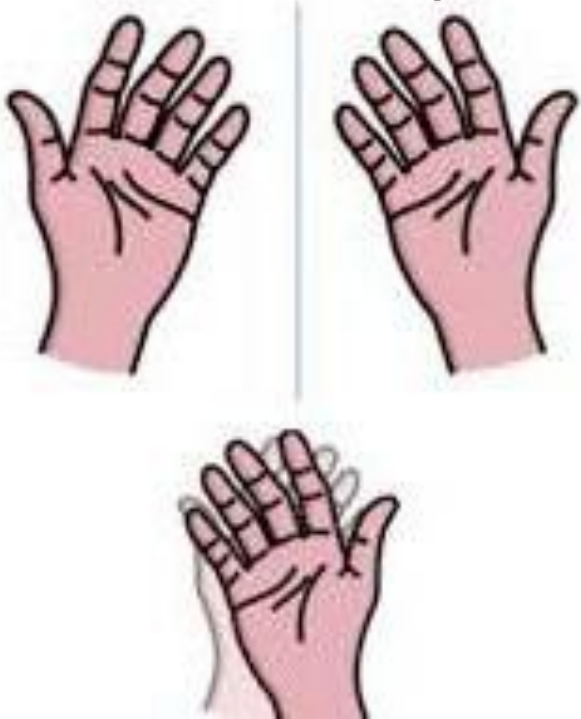
Connect chiral C with 4 different groups attached.



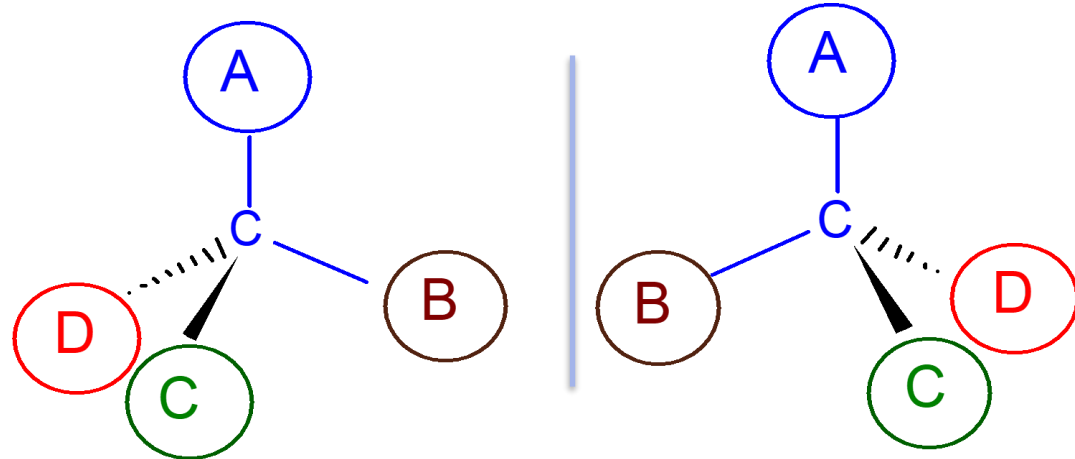
enantiomers

Enantiomers

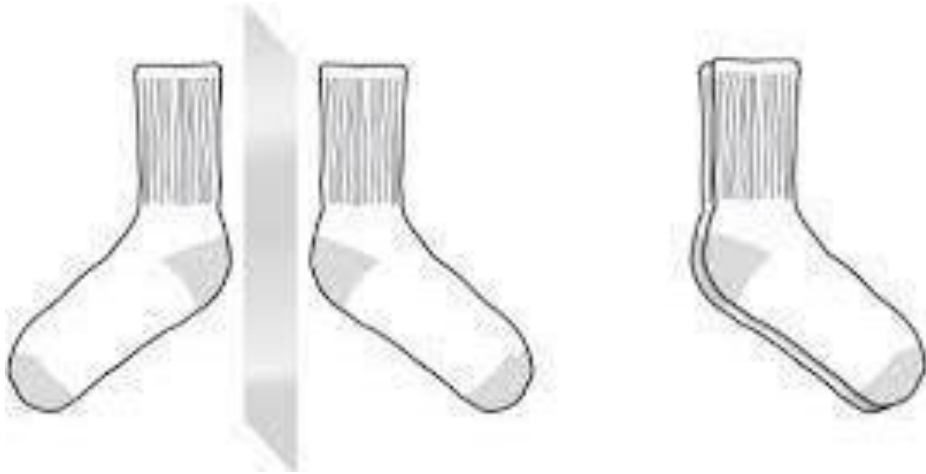
Mirror image



Mirror image



But cannot be superimposed



- Mirror image
 - Can be superimposed
- ➡ Not enantiomers



- Mirror image but cannot be superimposed
- ➡ Enantiomers

DIASTEREOMERS

- Diastereomers are stereoisomers that are not mirror images of one another and are non-superimposable on one another.
- Cis-trans isomers or geometrical isomers is a type of diastereomers
- Occurs only in alkenes and cyclic compounds.

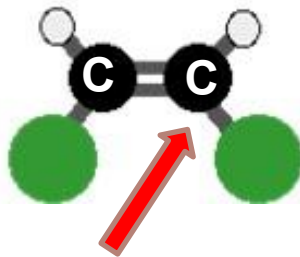
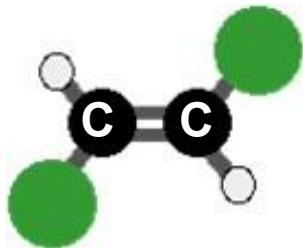
Requirements for cis-trans isomerism/geometrical isomerism:

1

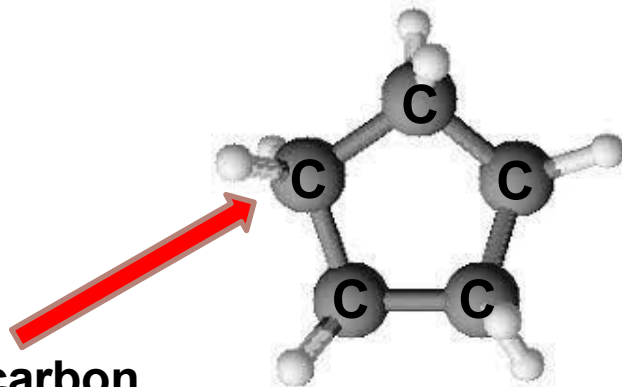
Restricted rotation about carbon-carbon double bond in alkenes or carbon-carbon single bond in cyclic compounds.

2

Each carbon atom of a site of restricted rotation has two different atoms or groups of atoms attached to it.



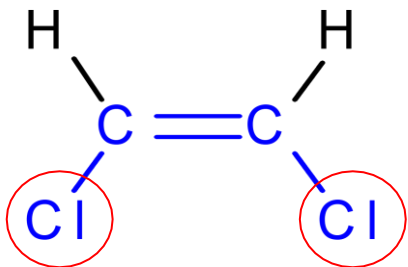
restricted rotation carbon



CIS-TRANS ISOMERS

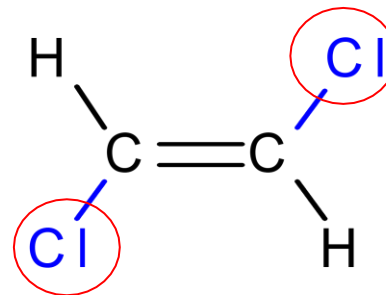
cis-isomers

same atom/group of atoms
are on the same side of the
site of restricted rotation



trans-isomers

same atom/group of atoms
are on the opposite side of
the site of restricted rotation

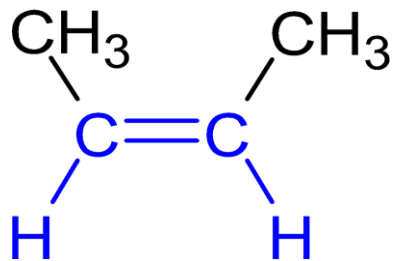


CIS-TRANS ISOMERS

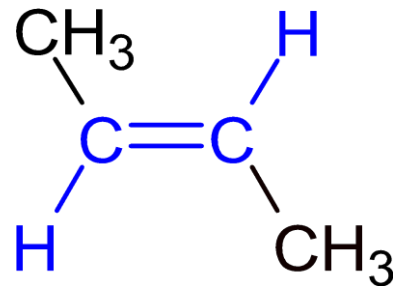
In alkene

- ✓ Restricted rotation about a carbon-carbon double bond in alkene.
- ✓ Different atoms or group of atoms attached to each carbon in C = C bond.

Example 1: 2-butene

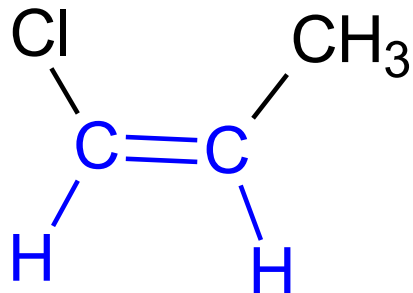


cis-2-butene

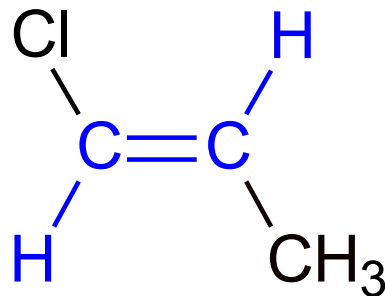


trans-2-butene

Example 2: 1-chloropropene

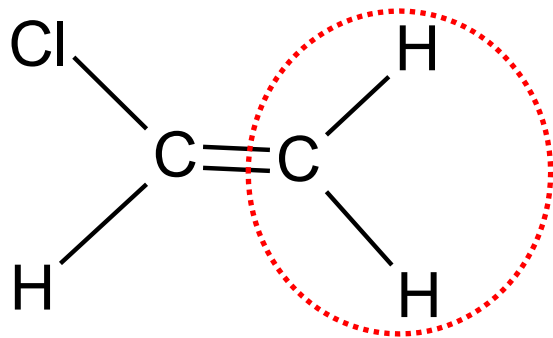


cis-1-chloropropene



trans-1-chloropropene

Example 3: chloroethene

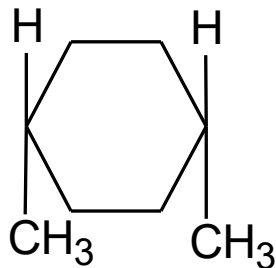


Does not exhibit cis-trans isomerism because identical atoms attach to carbon atom of the site restricted rotation.

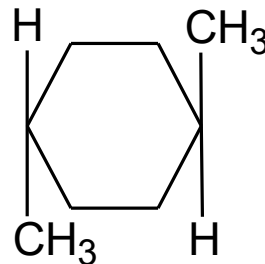
CIS-TRANS ISOMERS

In cycloalkane

Example : 1,4-dimethylcyclohexane



cis-1,4-dimethylcyclohexane



trans-1,4-dimethylcyclohexane

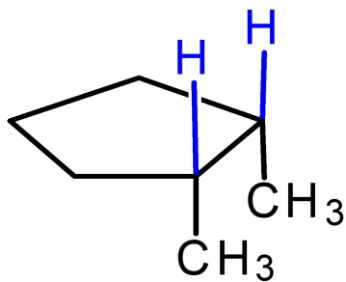
- ✓ Restricted rotation about a carbon-carbon single bond in cyclic compound.
- ✓ Different atoms or group of atoms attached to each carbon at site of restricted rotation.

Example:

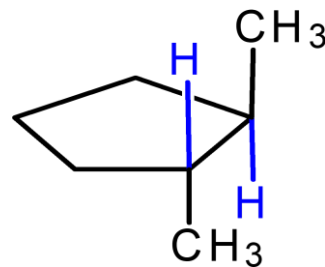
56

The molecular formula of the five-membered cyclic compounds is C_7H_{14} . Draw all geometrical isomers of the above five-membered ring compounds.

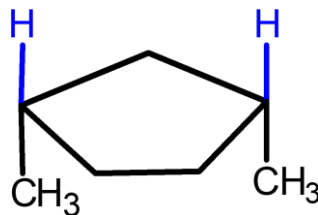
Answer:



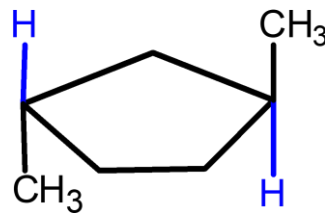
cis-isomers



trans-isomers



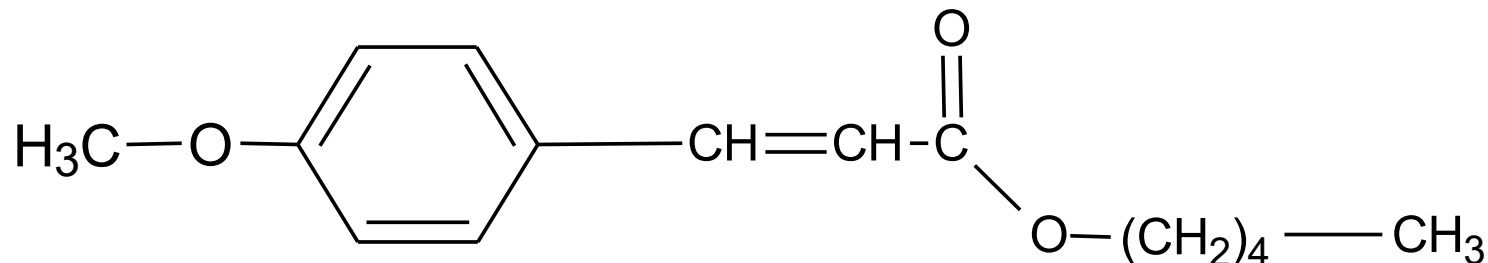
cis-isomers



trans-isomers

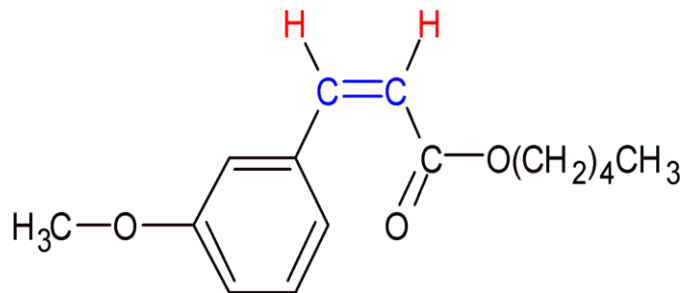
Exercise:

Given below is the structural formula of a compound found in a sunblock which protects skin from over-exposure to sunlight.

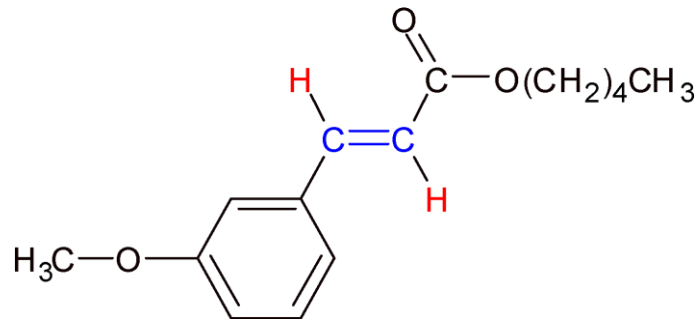


The compound has two isomers. Draw both the isomers. Name the type of isomerism.

Answer:



cis-isomers



trans-isomers

Type of isomerism : **cis-trans isomerism**

4.4 REACTIONS OF ORGANIC COMPOUNDS

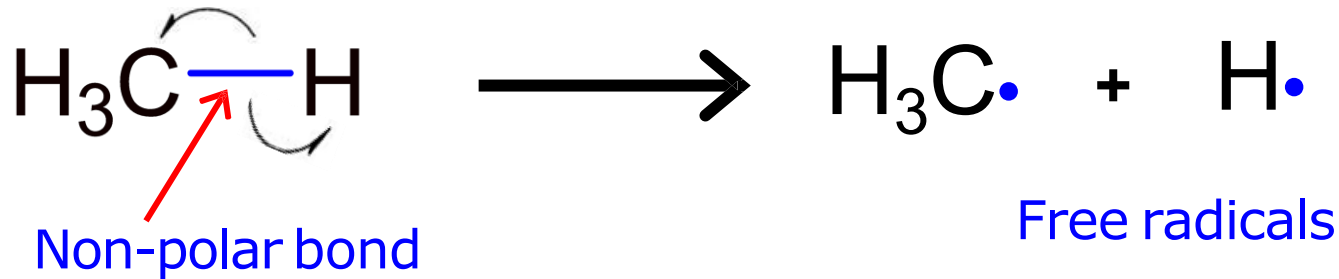
59

BOND CLEAVAGE

- All chemical reactions involve:
 - ✓ breaking of existing covalent bond in the reactants;
 - ✓ forming of new covalent bond in the products
- When a covalent bond breaks, the two bonding electrons are redistributed between two atoms.
- There are 2 types of bond cleavage:
 1. Homolytic cleavage
 2. Heterolytic cleavage

HOMOLYTIC CLEAVAGE

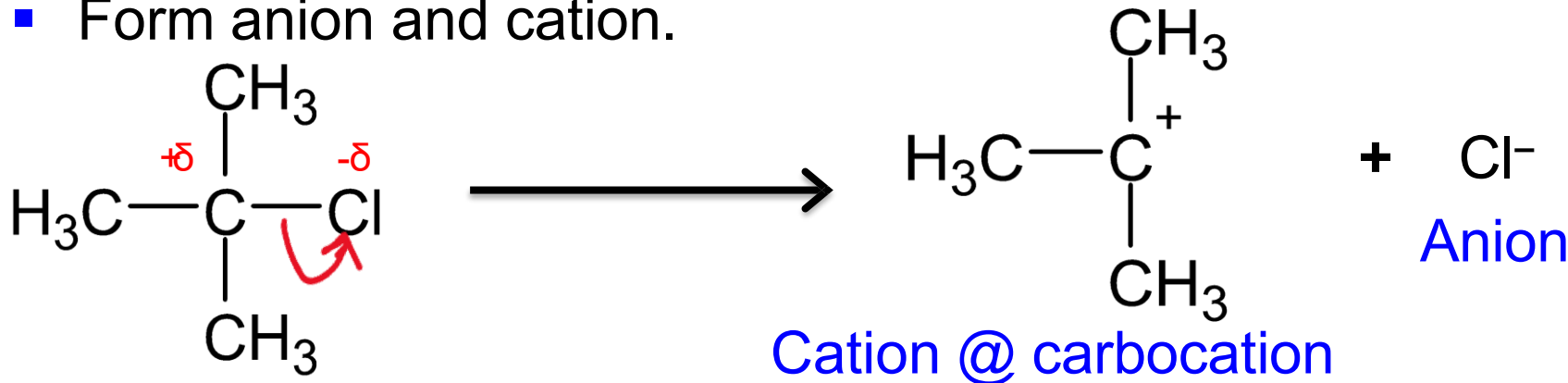
- Occurs in non-polar bond involving two atoms of similar electronegativity.
- Bonding electrons are equally distributed between the two bonded atom.
- Form free radicals.



: represents the transfer of one electron.

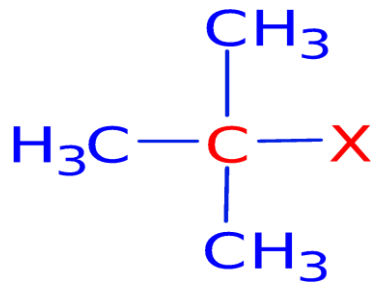
HETEROLYTIC CLEAVAGE

- Occurs in polar bond involving two atoms of different electronegativities.
- Bonding electrons are unequally distributed. Both electrons are transferred to the more electronegative atom.
- Form anion and cation.

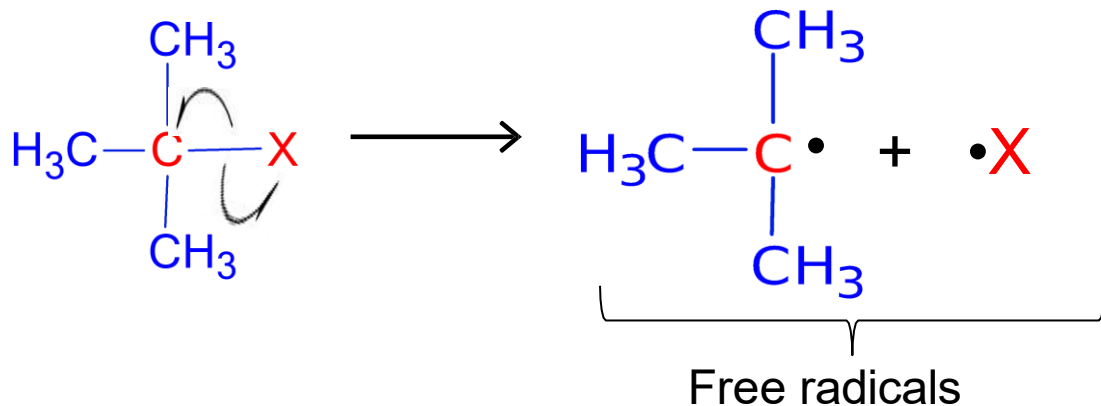


: represents the transfer of **two electrons** or **a pair of electrons**.

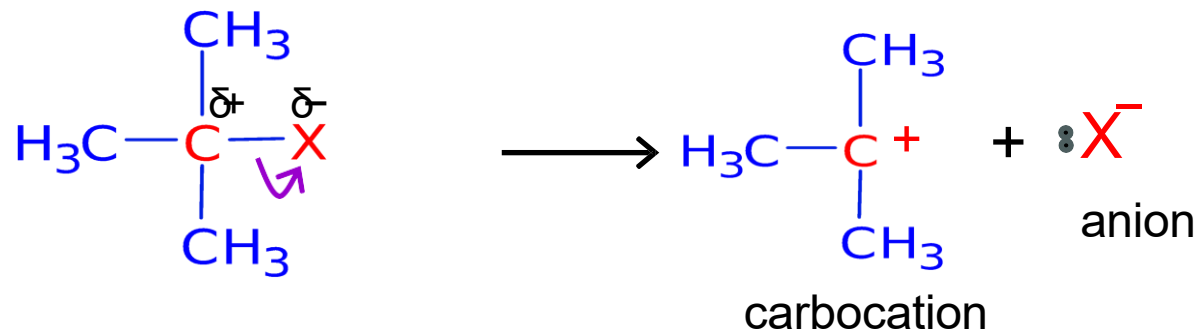
- 3 possible products formed from C—X bond cleavage:



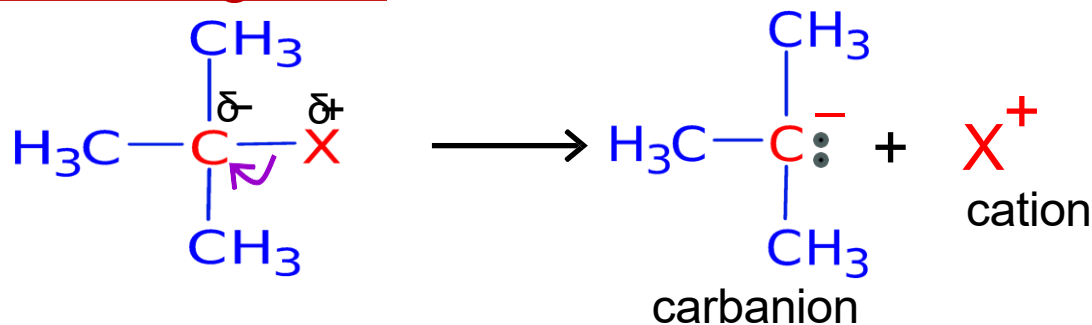
i. Free radical form via homolytic cleavage



ii. Carbocation form via heterolytic cleavage when atom X is more electronegative



iii. Carbanion via heterolytic cleavage when atom X is less electronegative



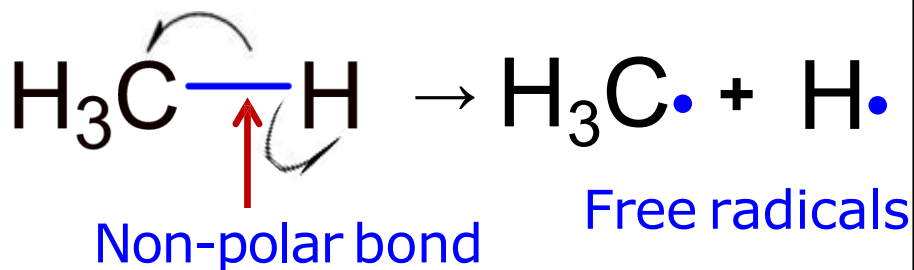
HOMOLYTIC CLEAVAGE vs. HETEROLYTIC CLEAVAGE

HOMOLYTIC

Occurs in non-polar bond involving equal sharing of electron pairs between two atoms of similar electronegativity.

A single bond breaks symmetrically into two equal part, leaving each atom with one unpaired electron.

Form free radical.

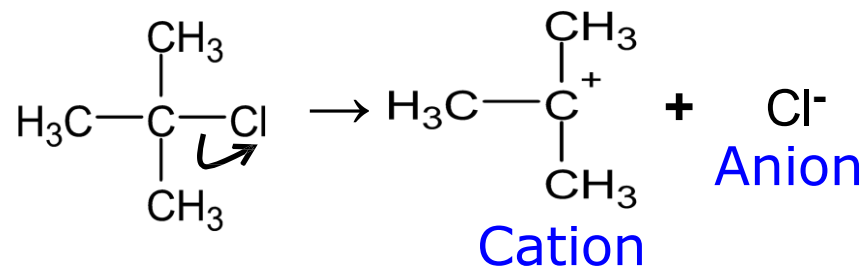


HETEROLYTIC

Occurs in polar bond involving unequal sharing of electron pairs between two atoms of different electronegativity.

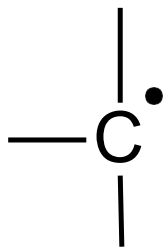
A single bond breaks unsymmetrically and both bonding electrons are transferred to the electronegative atom.

Form anion and cation.

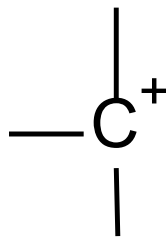


INTERMEDIATES IN ORGANIC REACTION

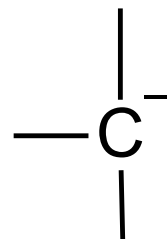
- Free radicals, carbocations and carbanions are intermediates in organic reaction.
- They are short-live and highly reactive.



Free radical



carbocation



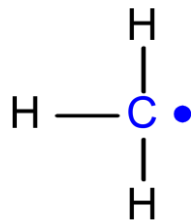
carbanion

INDUCTIVE EFFECT OF ALKYL GROUP ON THE INTERMEDIATE STABILITY

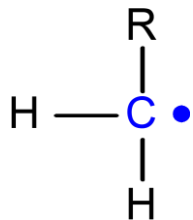
- Inductive effect is effect of an atom or group of atoms in pulling electrons towards itself or in pushing them away.
- Inductive effect will affect the stability of free radicals, carbocations and carbanions

CLASSIFICATION & STABILITY OF FREE RADICAL

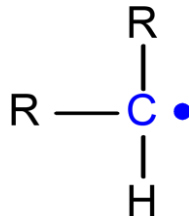
Increasing stability of free radical



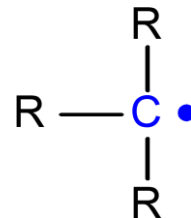
Methyl
free radical



primary
free radical



secondary
free radical



tertiary
free radical

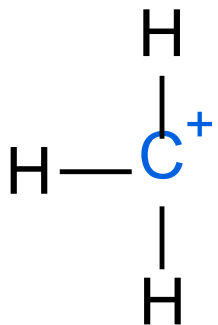


R is alkyl group which is electron-donating group (EDG)
that stabilizes the free radicals

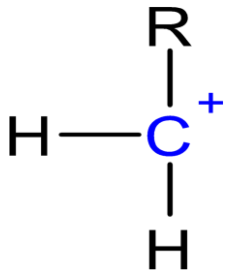


CLASSIFICATION & STABILITY OF CARBOCATION

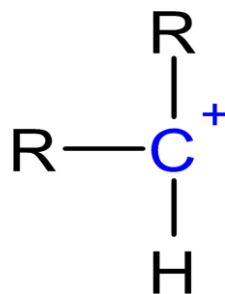
Increasing stability of carbocation



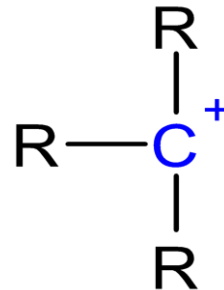
methylium
ion



primary
carbocation



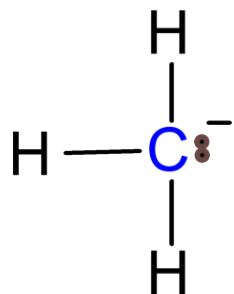
secondary
carbocation



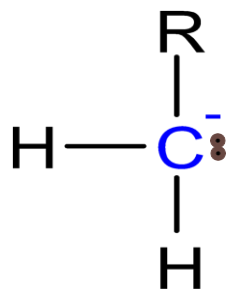
tertiary
carbocation

- Carbocations are **electron-deficient** species due to incomplete octet of the C atom.
- **Alkyl group (–R) is an electron-donating group** that **stabilises** the carbocation through inductive effect.
- Alkyl group increases the electrons density on the positively charged carbon.
- The higher the number of alkyl group, the more stable the carbocation

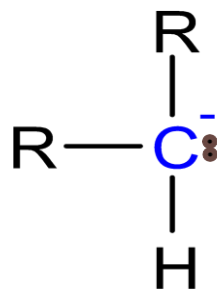
CLASSIFICATION & STABILITY OF CARBANION



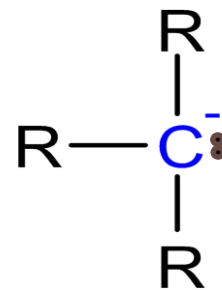
methyl
carbanion



primary
carbanion



secondary
carbanion



tertiary
carbanion

- Carbanions are high electron density species or **electron-rich species**.
- **Alkyl group** ($-R$) **is an electron-donating group** that **destabilises** the carbanion through inductive effect.
- Alkyl group increases the electron density on the negatively charged carbon.
- The lower the number of alkyl group, the more stable the carbanion

REAGENTS AND SITES OF ORGANIC REACTIONS

- A reagent is the chemical substance with which an organic compound reacts.
- There are 2 types of chemical reagents:

1 Electrophile

- An electron-deficient species that can accept a pair of electron to form a bond.
- attacks part of a molecule where the electron density is high (nucleophilic site)

2 Nucleophile

- An electron-rich species that can donate a pair of electron to form a bond.
- attacks part of a molecule where the electron density is low (electrophilic site)

Electrophile

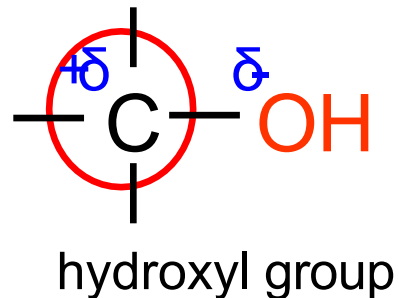
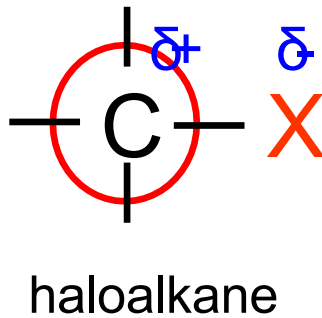
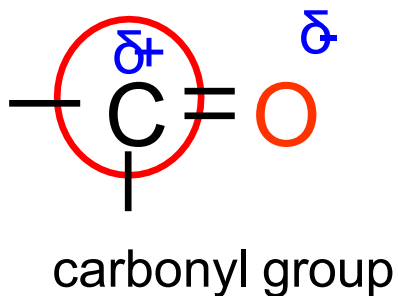
- Examples:
- **Cations**
 - **Carbocation**
 - **Lewis acids :**
A species that can accept electron pair to form a covalent bond such as AlCl_3 , FeCl_3 , BF_3

Nucleophile

- **Anions**
- **Carbanion**
- **Lewis bases :**
species which can donate lone pair electrons such as NH_3 , H_2O

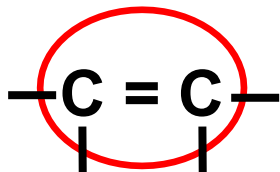
Electrophilic Sites in Organic Molecules

The site of a molecule with **low electron density** around a polar bond such as:

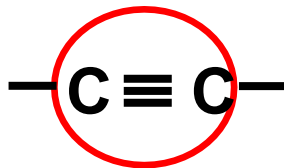


Nucleophilic Sites in Organic Molecules

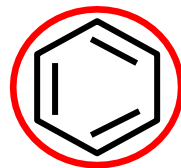
The site of a molecules with **high electron density** around the C-C multiple bond:



Alkenes



Alkynes



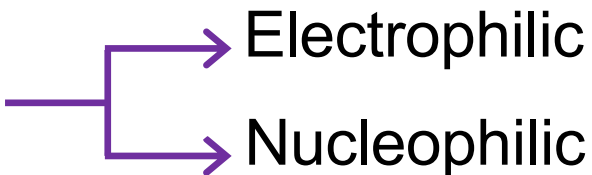
Aromatic
compounds

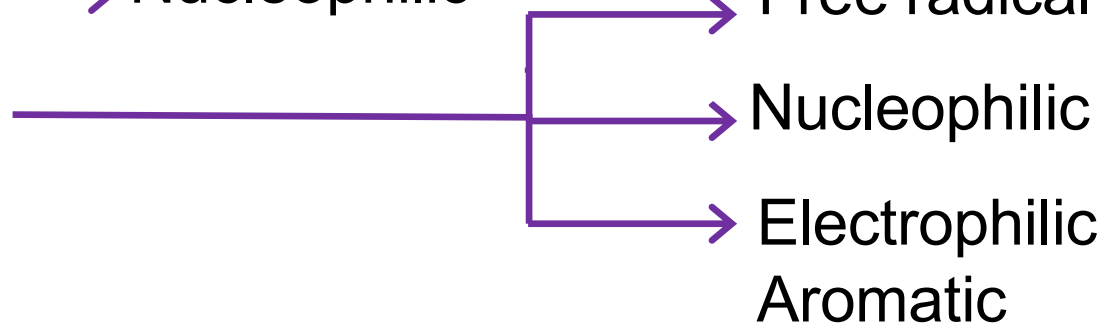
TYPES OF CHEMICAL REACTION

76

- There are 4 main types of reaction in organic chemistry .

- ☐ **Rearrangement**

- ☐ **Addition** 
 - Electrophilic
 - Nucleophilic

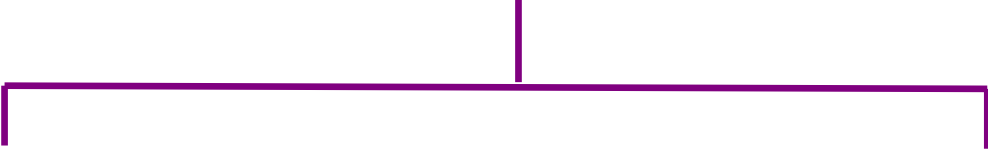
- ☐ **Substitution** 
 - Free radical
 - Nucleophilic
 - Electrophilic Aromatic

- ☐ **Elimination**

ADDITION REACTION

- Characteristic reaction of compounds with multiple bond.
- Atoms or groups of atom are added to adjacent atoms of a multiple bond.
- All parts of the adding reagent appear in the product; two molecules become one.
- Involve in breaking of one π bond to form two σ bonds.

There are 2 types of addition reactions:



Electrophilic Addition

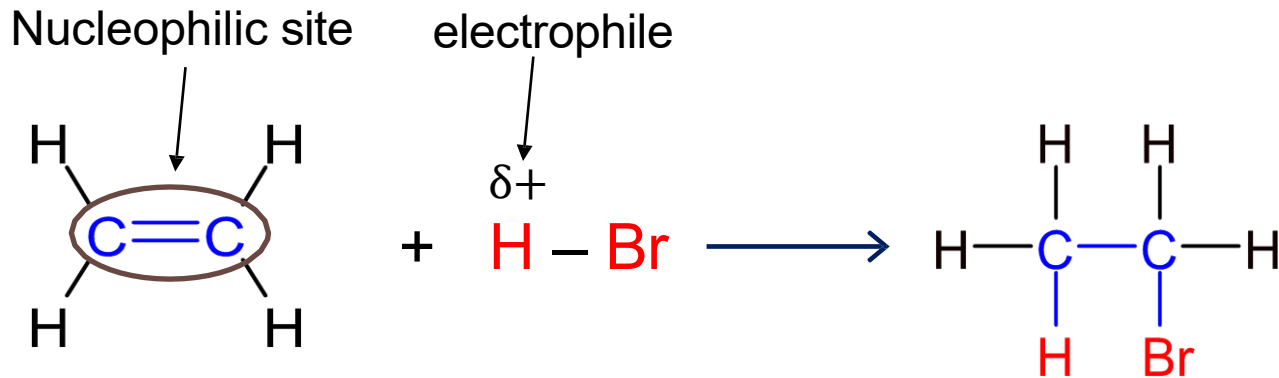
- Initiated by electrophile
- Typical reaction of alkene

Nucleophilic Addition

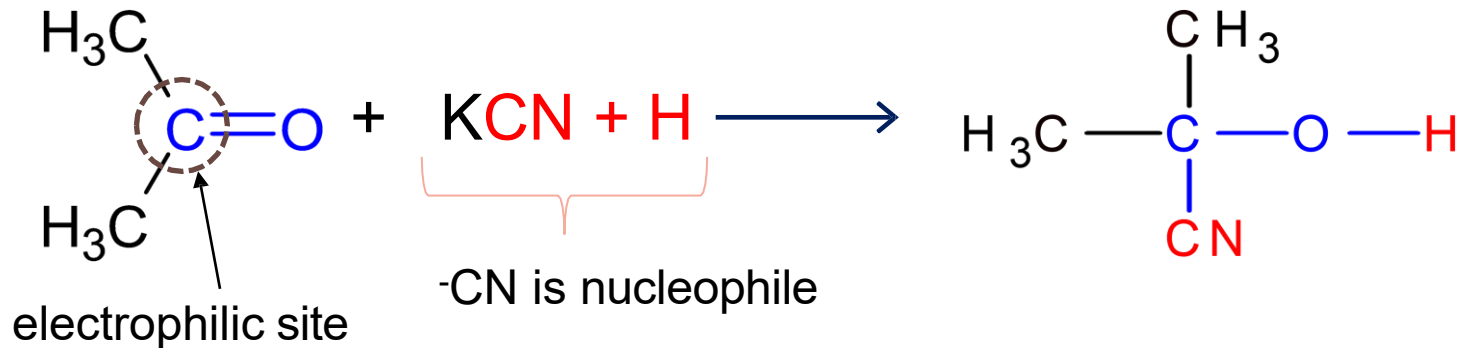
- Initiated by nucleophile
- Typical reaction of carbonyl compounds

Examples:

Electrophilic addition



Nucleophilic addition



ELIMINATION REACTION

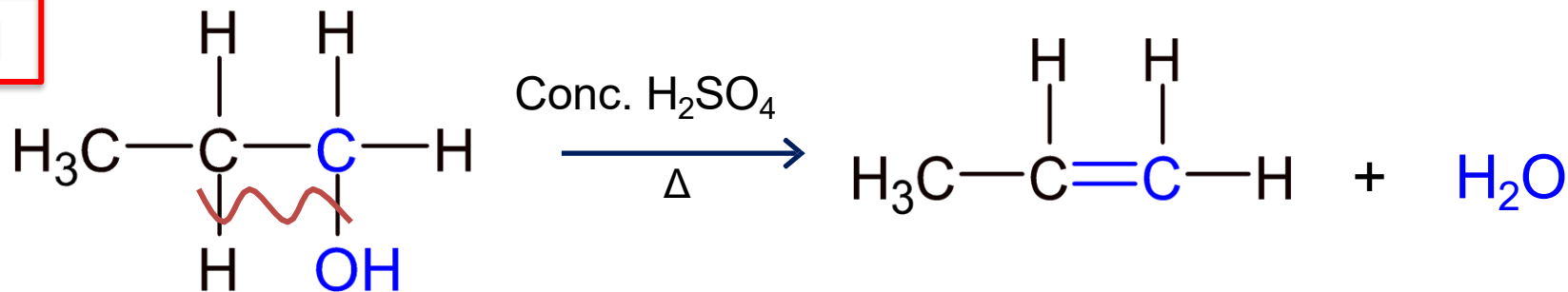
- The opposite of an addition reaction.
- Atoms or groups of atoms are removed from adjacent C atom.
- Results in formation of unsaturated molecules (molecules with double or triple bond).
- Involve in breaking of two σ bonds to form one π bond.



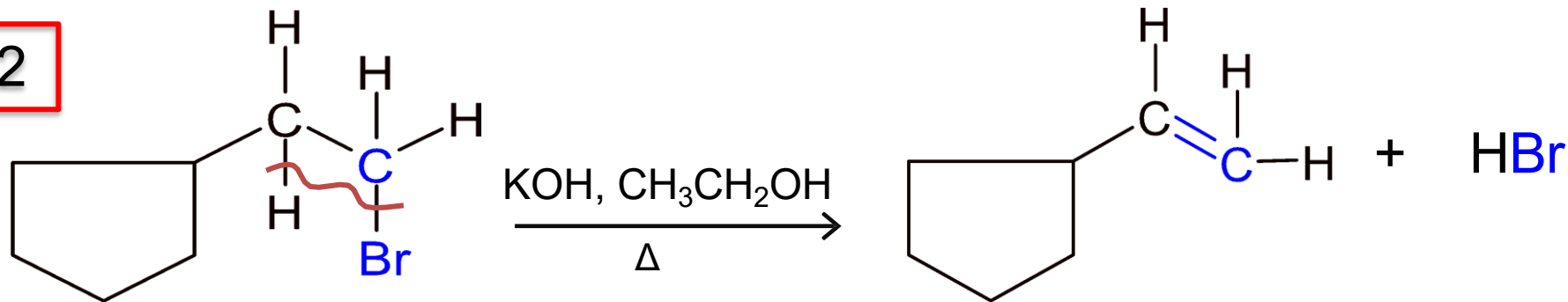
Example:

81

1

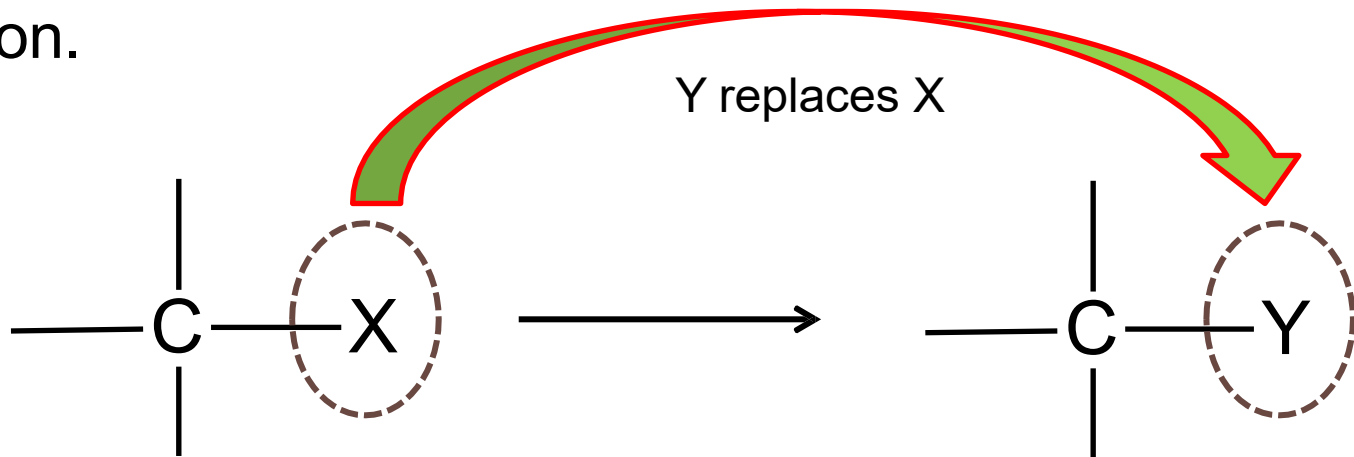


2



SUBSTITUTION REACTION

- Characteristic reaction of **alkanes, haloalkanes and aromatic compounds**.
- One atom or group of atoms is substituted by another atom or group of atoms in a molecule.
- Saturated reactant remain as saturated product after substitution.



There are 3 types of substitution reactions:

83

Free radical substitution

- Typical reaction of alkane.
- Involving free radical as intermediates.

Electrophilic substitution

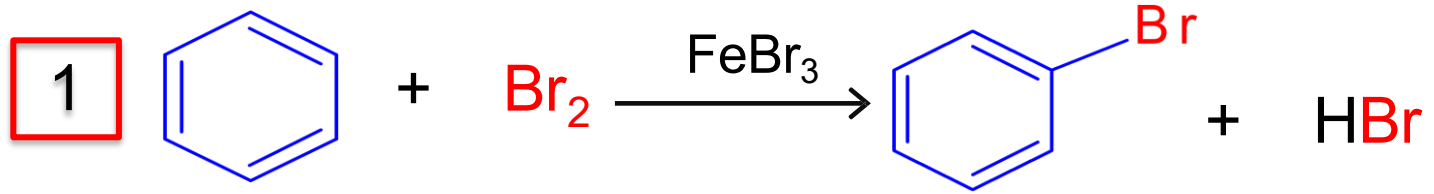
- Typical reaction of aromatic compounds (e.g:benzene, toluene)
- The nucleophilic site of a molecule is attacked by an electrophile.

Nucleophilic substitution

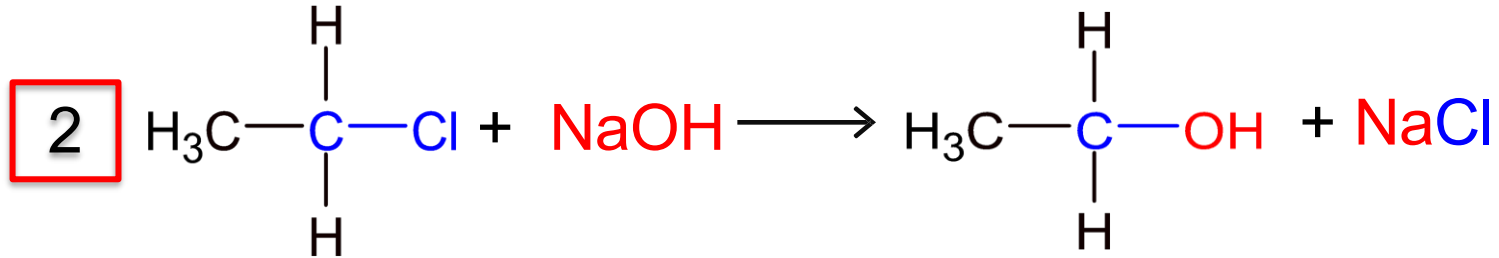
- Typical reaction of compounds with polar bond (e.g. : haloalkane & alcohol)
- The electrophilic site of a molecule is attacked by a nucleophile.

Examples:

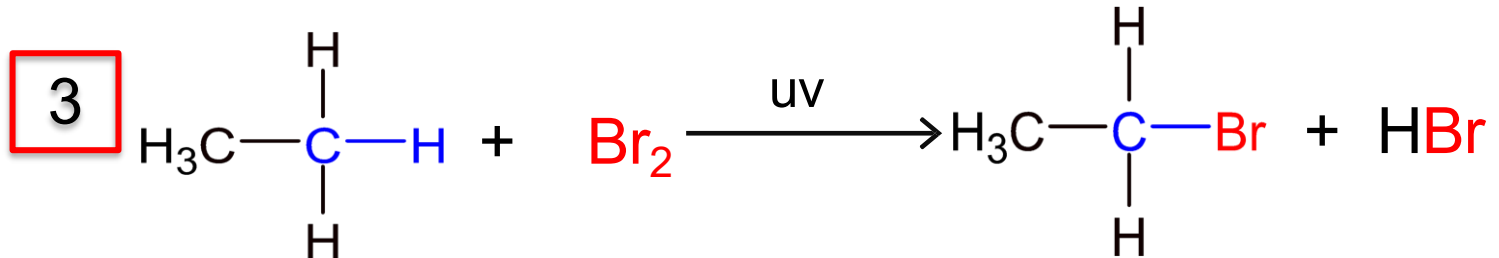
**Electrophilic
aromatic
substitution**



**Nucleophilic
substitution**

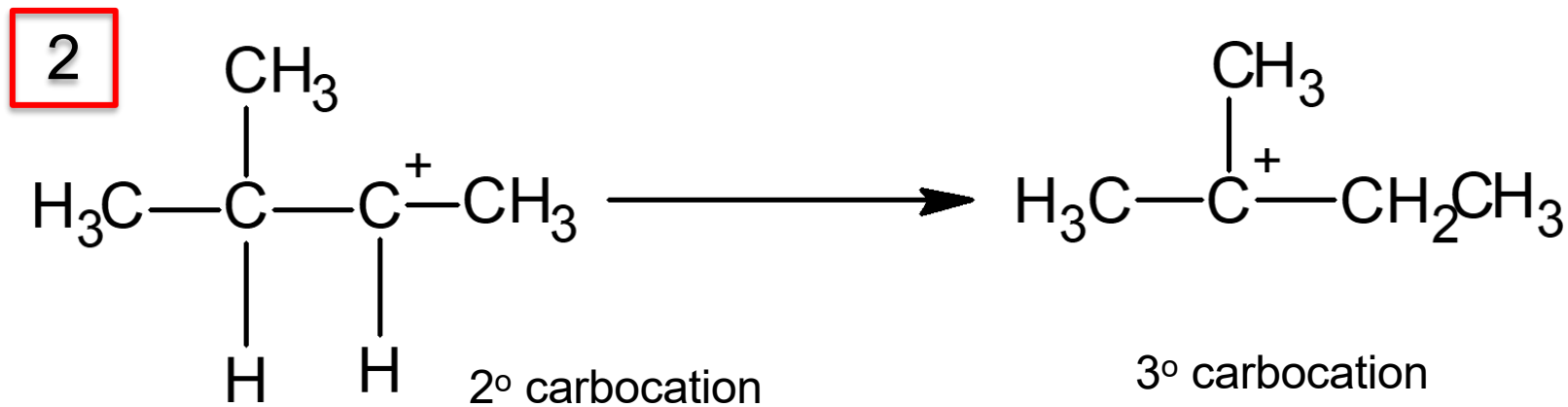
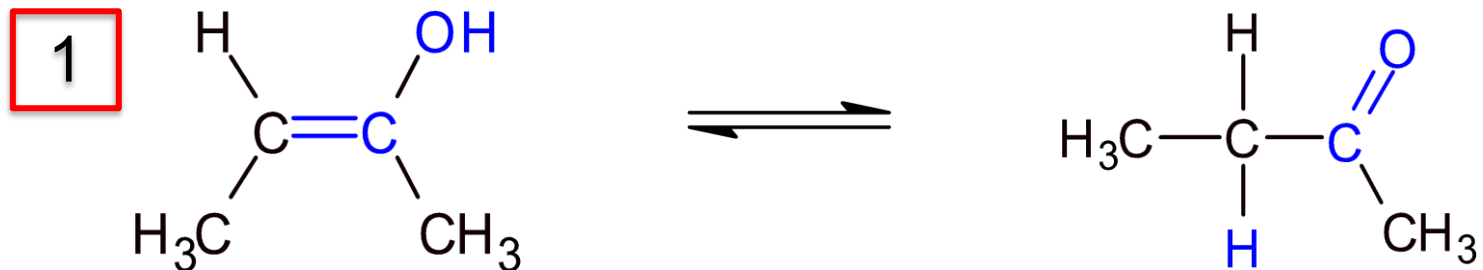


**Free radical
substitution**



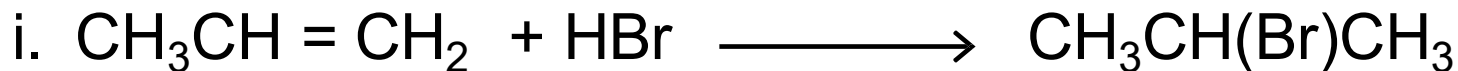
REARRANGEMENT REACTION

- Atoms or groups of atoms change position within a molecule (atoms @ groups of atoms and/or bonds are re-organised).



Exercise:

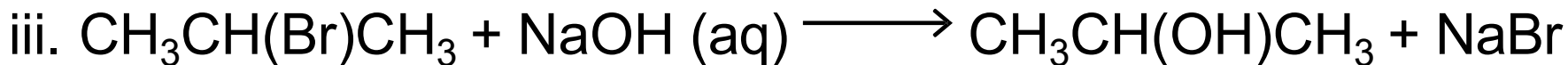
State the type of reaction for each of the following reactions :



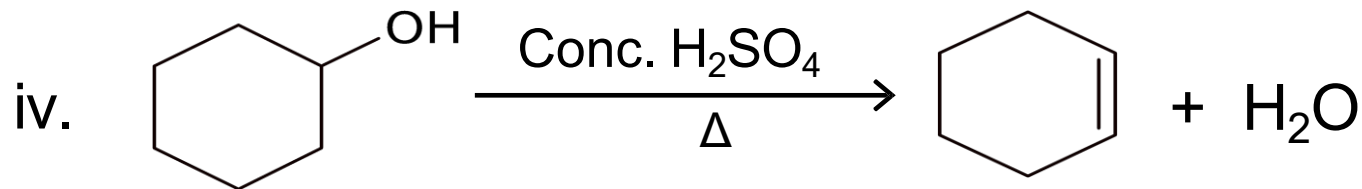
⇒ Electrophilic addition reaction



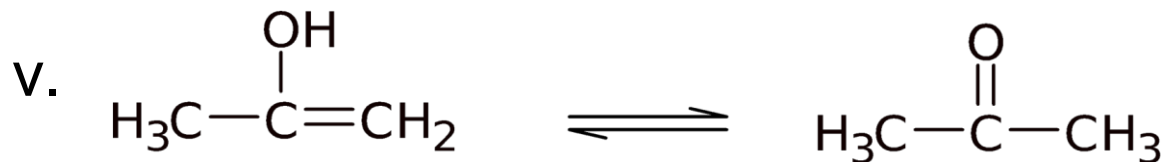
⇒ Free radical substitution reaction



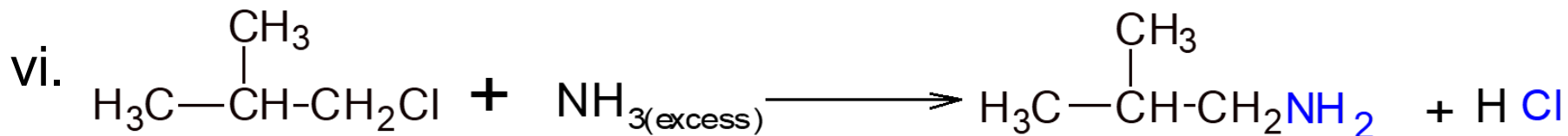
⇒ Nucleophilic substitution reaction



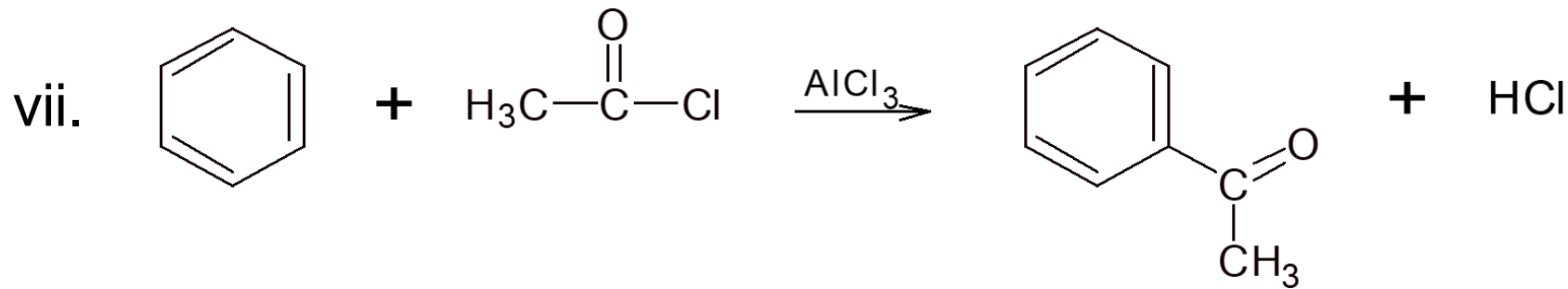
⇒ Elimination reaction



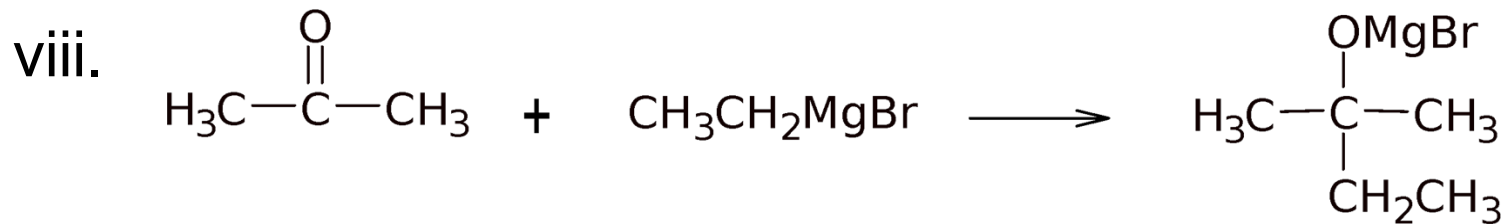
⇒ Rearrangement



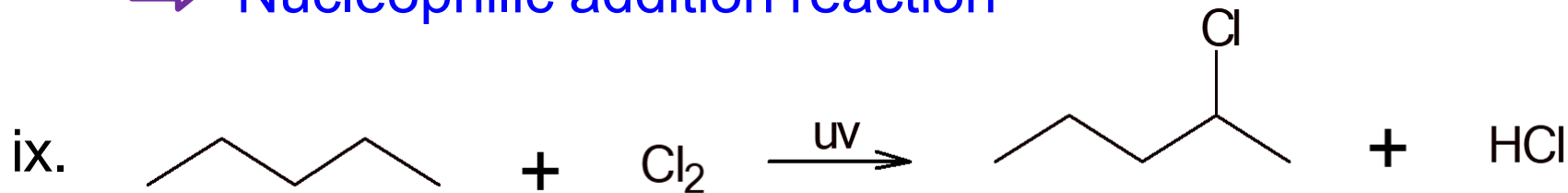
⇒ Nucleophilic substitution reaction



⇒ Electrophilic aromatic substitution reaction



⇒ Nucleophilic addition reaction



⇒ Free radical substitution reaction

The slide features decorative geometric shapes in the corners. The top-left corner is filled with overlapping triangles in shades of blue, green, and red. The bottom-right corner features a cluster of overlapping triangles in various shades of gray.

THE END